ECERC Revisited: Final Report

Divyansh Dubey 12340690
Keshav Mishra 12341140
Lakshay Gupta 12341300

Oct 30, 2026

Abstract

We extended ECERC, a multimodal model for emotion recognition in conversation, on three limitations: category confusion, class imbalance and cause gating. We beat the authors' checkpoints on both datasets, in two separate ways. In Approach A, merged per dataset, each teammate merged Confidence-Aware Cause Gating (CACG) into their own pipeline and retrained on their own dataset. In Approach B, all three limitations were merged into one rewritten codebase and retrained on a laptop. The final systems reach 72.43 to 72.77 weighted F1 (wF1) on IEMOCAP against 71.78, and 53.77 to 53.82 macro-F1 (mF1) on MELD against 50.87, and the two merged systems agree on what works.

Better inputs produced the accuracy gains, not new losses. WavLM audio drove the IEMOCAP gain, and zero-shot LLM fusion drove the MELD gain. Across Limitations 1 and 2, 17 earlier methods fell short of their goal, and those failures pointed to the methods that worked. CACG added no proven accuracy, but it makes cause weighting readable. The gaps between Approaches A and B are too small to rank them. Approach A's IEMOCAP result has stronger evidence, with 20 seeds against 5.

1 Introduction

ECERC (Zhang and Tan, ACL 2025) [1] recognises emotions in conversation by combining text, audio and visual evidence with four causes: the speaker's own past emotion, other speakers' emotions, the utterance's own event, and other speakers' events. The team kept ECERC's structure and addressed one failure mode per person.

Limitation	Owner	Dataset	Problem
L1: category confusion	Divyansh Dubey	IEMOCAP	Happy is confused with Excited, and Angry with Frustrated
L2: class imbalance	Keshav Mishra	MELD	Fear and Disgust are under 6% of the data and score poorly
L3: cause gating	Lakshay Gupta	Both	Causes are weighted the same way whatever the evidence, and predictions are over-confident

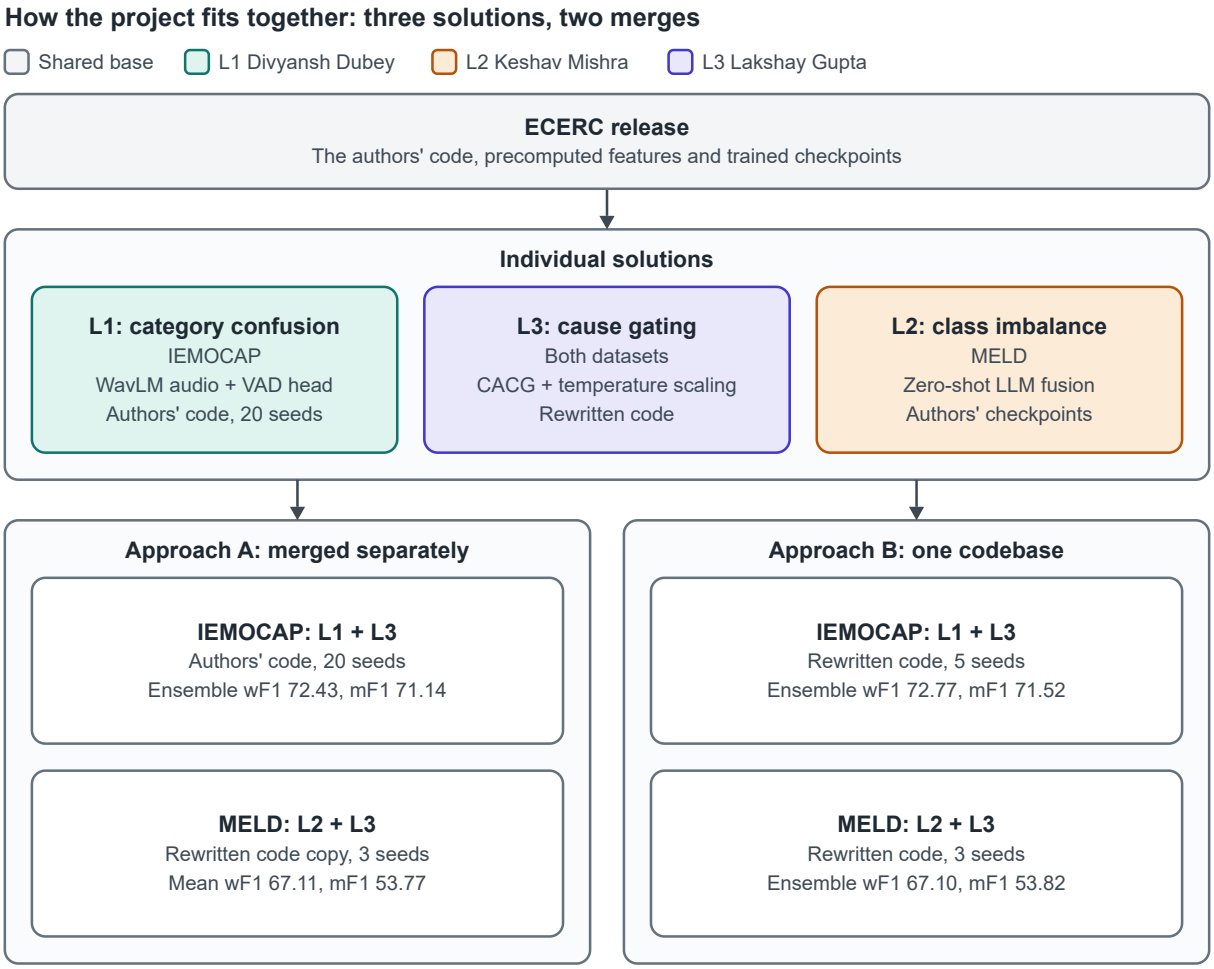


Figure 1: How the individual solutions feed the two merged systems

Two approaches. The team built two merged systems. In Approach A, each dataset was merged separately: Divyansh added CACG to the authors' IEMOCAP code, and Keshav added L2 to a copy of the rewritten code from L3. In Approach B, all three limitations sit in one rewritten codebase behind command-line flags, and every run was retrained on one laptop.

Contributions. This report makes the following contributions:

- A reproduction of ECERC on both datasets, and a rewritten codebase verified as equivalent to the authors' model (Sections 3.1 and 5.2).
- WavLM audio with VAD supervision for category confusion on IEMOCAP (L1, Section 3.2).
- Zero-shot LLM fusion for class imbalance on MELD (L2, Section 3.3).
- Confidence-Aware Cause Gating (CACG) with temperature scaling for cause gating and calibration (L3, Section 3.4).
- Two merged systems, one per dataset and one unified codebase, that both beat the authors' checkpoints (Section 3.5), with a record of all 31 methods tried across the three limitations (Section 6).

Each member's individual contribution is listed in Appendix A.

2 Related Work / Literature Review

2.1 Emotion recognition in conversation

Emotion recognition in conversation (ERC) assigns an emotion to every utterance in a dialogue, using the utterance itself and the turns before it. Earlier models track each speaker's state with recurrent networks, as in DialogueRNN [4], or treat the dialogue as a graph of utterances, as in DialogueGCN [5] and its multimodal extension MMGCN [6]. ECERC [1] instead separates evidence from causes: it gates the text, audio and visual evidence of each utterance, then attends over the dialogue history through four cause channels before classifying. IEMOCAP [2] and MELD [3] are the two benchmarks reported in the ECERC paper, and the two used here.

2.2 Speech and text representations

ECERC's IEMOCAP audio is the 1,582-d IS10 feature set from the Interspeech 2010 Paralinguistic Challenge [7], a fixed set of hand-designed acoustic descriptors. Self-supervised speech models such as WavLM [8] learn representations directly from raw audio, and Whisper [9] is trained for speech recognition on large weakly supervised data. L1 replaces IS10 with WavLM, and L2 tested Whisper-large-v3. ECERC's text and semantic features come from RoBERTa [10]. IEMOCAP also provides dimensional valence, arousal and dominance (VAD) ratings and individual annotator votes, which L1 uses as auxiliary targets and as soft labels [11].

2.3 Class imbalance

Common remedies for rare classes change the loss, the sampling or the decision rule. Focal loss [12] down-weights easy examples, and ECERC's own MELD loss is already focal. Class-balanced loss [13] re-weights classes by their effective number of samples, and post-hoc logit adjustment [14] shifts the decision rule by the class prior. Supervised contrastive learning [15] pulls same-class representations together. L2 tested all of these families. Large instruction-tuned language models such as Qwen2.5 [16] can label text without task-specific training, which is the basis of L2's zero-shot fusion, and knowledge distillation [17] is the standard way to move such a model's knowledge into a smaller one.

2.4 Calibration and uncertainty

Modern neural networks tend to be over-confident, and temperature scaling, a single parameter fitted on validation data, is a strong and simple fix [18]. Expected calibration error (ECE) [19] measures the gap between confidence and accuracy. Ensembles of independently trained models improve both accuracy and uncertainty estimates [20], and selective classification [21] holds back predictions whose confidence is low. L3 uses all four.

Position of this work. Rather than proposing a new architecture, this project keeps ECERC and asks which of these remedies fixes each of its three failure modes. The answer differs from the usual expectation: on both datasets, the gains came from better evidence (WavLM audio and a zero-shot LLM), not from loss, sampling or decision-rule changes.

3 Proposed Methodology

3.1 Base model: ECERC and the rewritten code

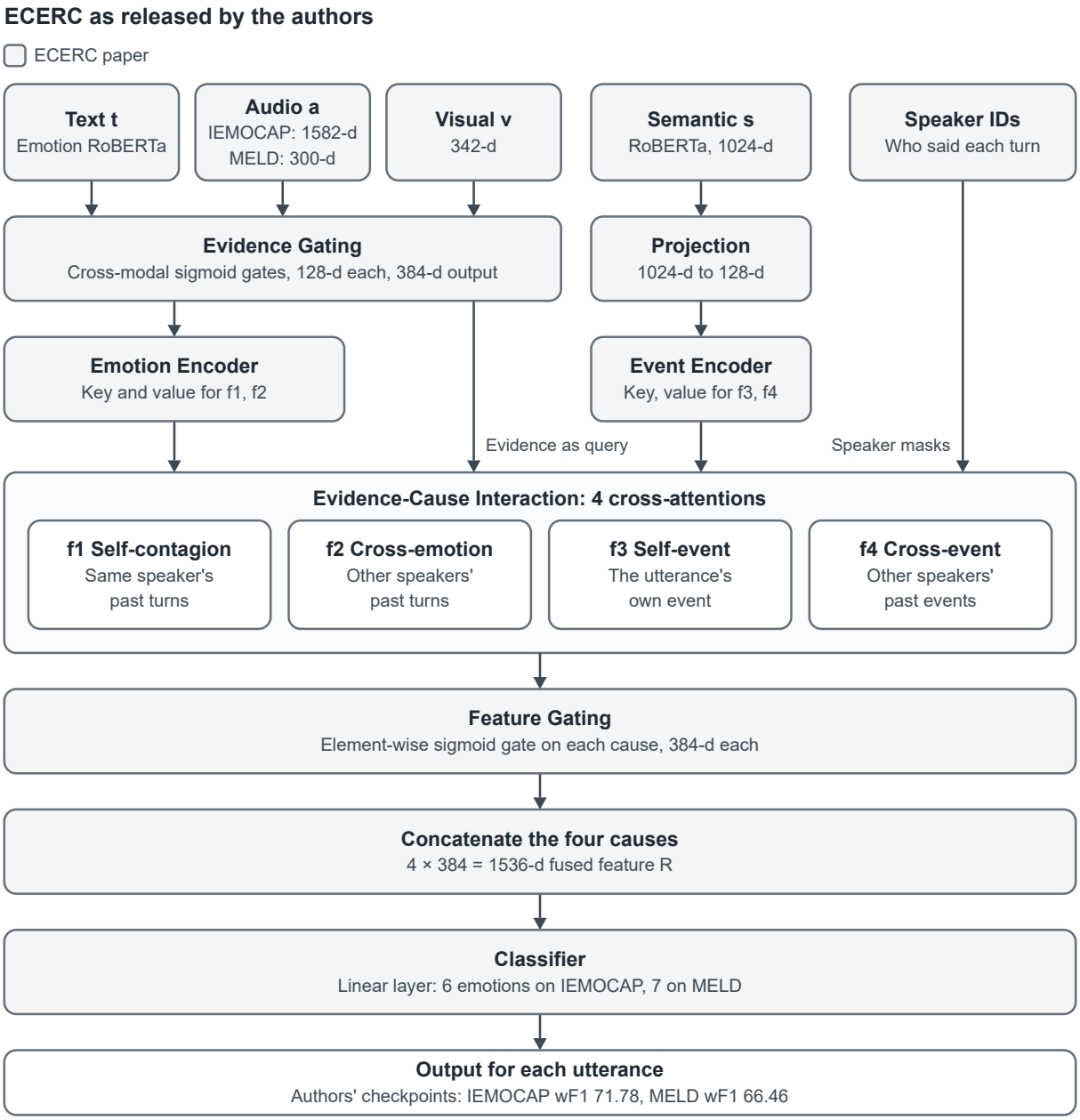


Figure 2: ECERC as released by the authors

The rewrite. The rewritten code puts both datasets in one package, keeps the authors' parameter names and adds a flag for every extension. With the same weights and inputs, it gives the same outputs as the official model, with a maximum difference of 0.0. The authors' checkpoints load into it unchanged. It changes how the code is organised, not what the model computes. The rewrite was verified as equivalent to the authors' model, so it is a correct base to build on, not an accuracy improvement.

Problems found in the released code. We kept all three, so that the baseline stays comparable with the authors' model.

- **No cause-level softmax.** The project proposal described a softmax over the four causes, but the release uses an element-wise sigmoid gate on each one.
- **Empty-history attention.** A cause with no valid history gets uniform attention over the whole padded dialogue, including future turns. This affects 28.6% (same speaker) and 15.8% (other speakers) of MELD test utterances, and 3 to 4% on IEMOCAP.
- **MELD modality routing.** The MELD loader swaps the audio and visual slots.

3.2 Limitation 1: WavLM audio and VAD supervision (Divyansh Dubey, IEMOCAP)

Five training changes on the original features failed to fix the confused pairs (Section 6.1), which showed the real problem was missing information. The final method therefore adds better audio.

WavLM-large [8] (layer 20, averaged over time) gives a 1,024-d vector that replaces the 1,582-d IS10 audio features for all 7,433 utterances. Text, video and the model stay the same. A training-only head predicts the human valence, arousal and dominance ratings with loss weight 1.0.

$$L = w_y \,\mathrm{CE}(p, y) + \lambda_{vad}\,\mathrm{MSE}\big(\mathrm{head}(R),\ \mathrm{zscore}(\mathrm{VAD})\big), \qquad \lambda_{vad} = 1$$

An audio-only check confirmed the potential before full training, and it predicted the final pattern.

Audio-only classifier	IS10	WavLM
6-class macro-F1	34.8	51.8
Angry vs Frustrated, balanced accuracy	60.5%	72.8%
Happy vs Excited, balanced accuracy	64.2%	66.5%

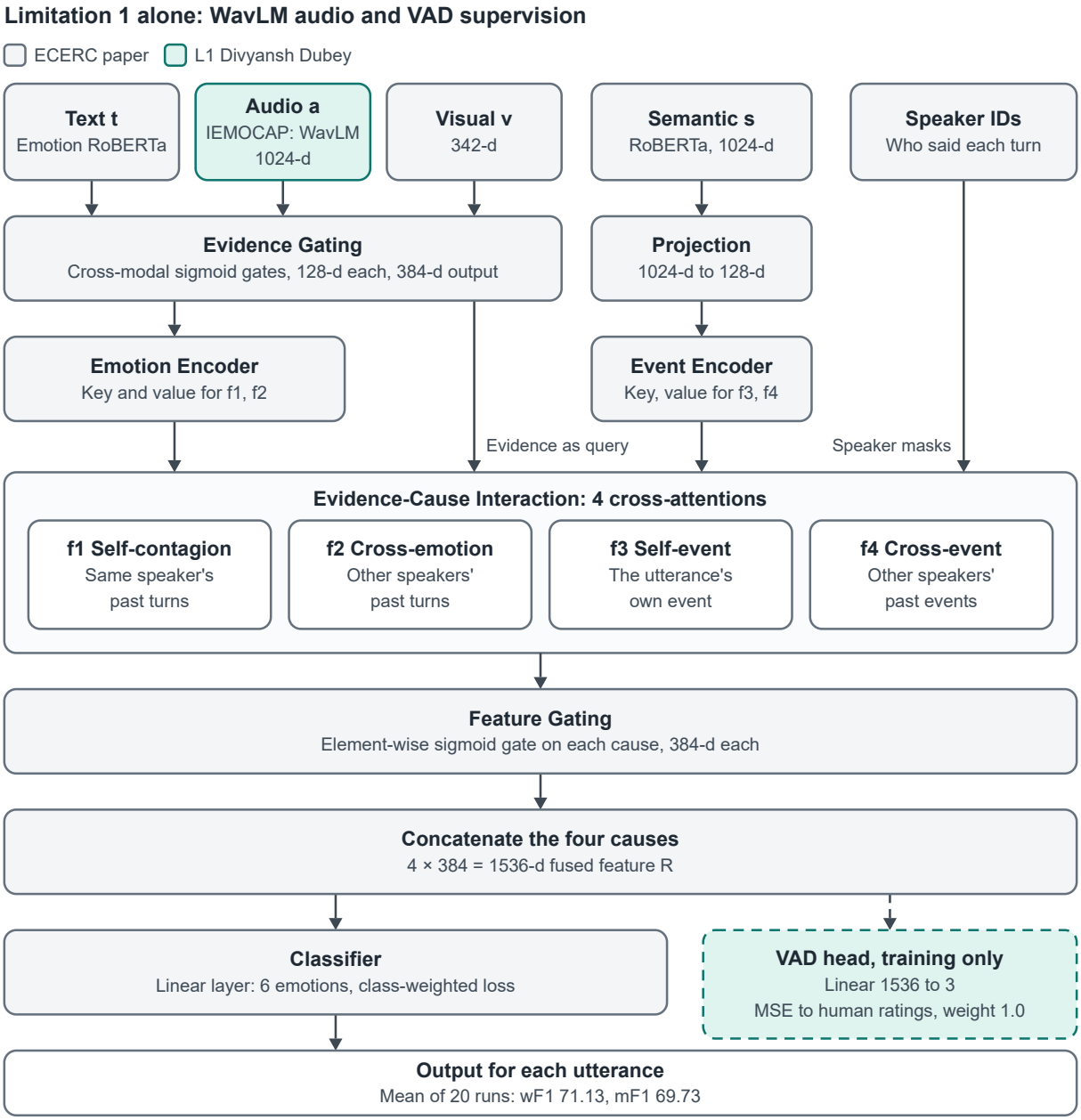


Figure 3: Limitation 1: WavLM audio and a training-only VAD head

3.3 Limitation 2: zero-shot LLM fusion (Keshav Mishra, MELD)

Twelve methods gave no confirmed macro-F1 gain (Section 6.2), which showed the limit is ECERC's evidence, not its loss, decision rule or training budget. The final method therefore adds a second, independent source of evidence.

Qwen2.5-32B-Instruct [16] (4-bit, frozen) scores every utterance from the dialogue so far. It is never trained on MELD. Its scores are calibrated without labels, using its mean dev prediction q and the training prior π, and then combined with ECERC's after training. The weight $w = 0.2$ was chosen on dev.

$$\tilde{p}_{LLM}(c) \propto \frac{p_{LLM}(c)}{q(c)}\,\pi_{train}(c) \qquad \log p_{final} = (1-w)\log p_{ECERC} + w \log \tilde{p}_{LLM}$$

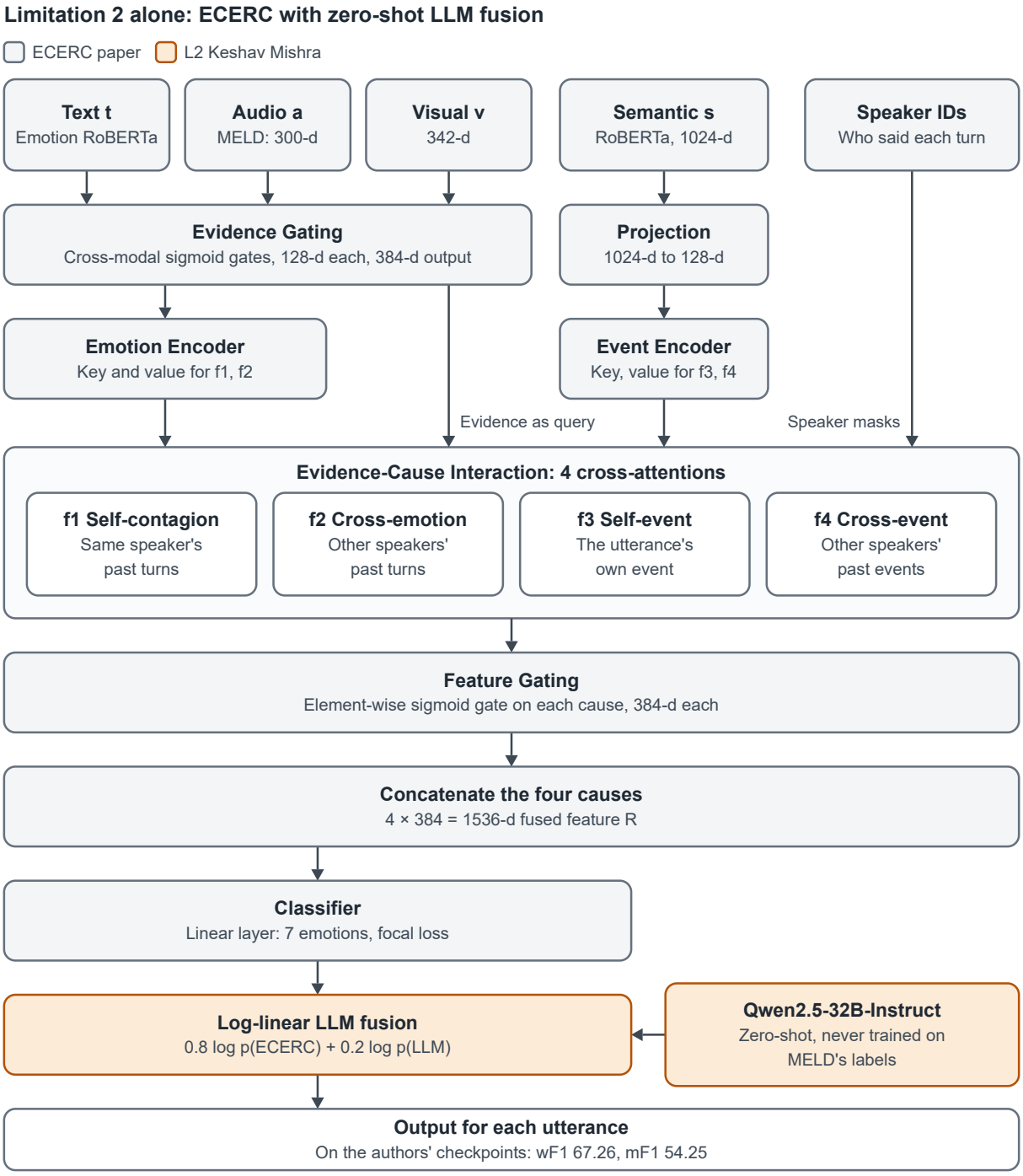


Figure 4: Limitation 2: ECERC with zero-shot LLM fusion

3.4 Limitation 3: Confidence-Aware Cause Gating (Lakshay Gupta, both datasets)

CACG sits after ECERC's Feature Gating and adds a distribution over the four causes for each utterance. Its parameters add 0.11% to ECERC's size. With the mask off, it starts identical to ECERC, which the tests confirm.

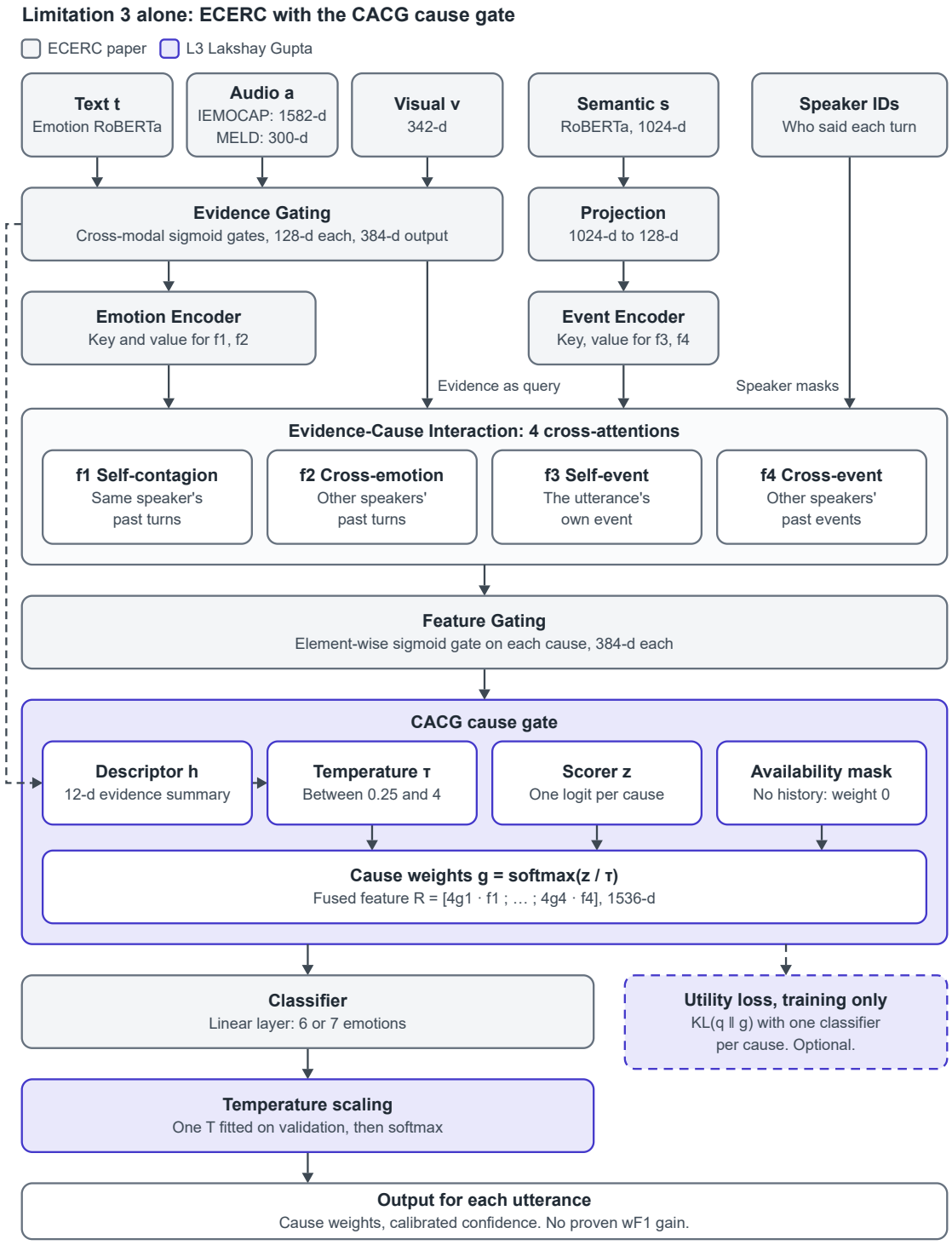


Figure 5: Limitation 3: the CACG cause gate and temperature scaling

Evidence descriptor. A 12-d summary holds 3 modality-gate means, 3 history counts and 6 pairwise cosines between cause vectors.

$$h_i = [\,\bar{w}_i^t,\ \bar{w}_i^a,\ \bar{w}_i^v,\ \log(1+n_i),\ \log(1+n_i^{same}),\ \log(1+n_i^{other}),\ \cos(\tilde{f}_{ia}, \tilde{f}_{ib})_{a<b}\,]$$

Temperature and masked weights. A small network predicts τ from h. A linear scorer gives one

logit per cause, and m removes causes with no history.

$$\tau_i = 0.25 + 3.75 \cdot \sigma\big(\mathrm{MLP}(h_i)\big) \qquad g_{ik} = \frac{m_{ik}\exp(z_{ik}/\tau_i)}{\sum_j m_{ij}\exp(z_{ij}/\tau_i)}$$

Fusion. Multiplying by 4 means a uniform gate leaves the features unchanged.

$$R_i = [\,4g_{i1}\tilde{f}_{i1};\ 4g_{i2}\tilde{f}_{i2};\ 4g_{i3}\tilde{f}_{i3};\ 4g_{i4}\tilde{f}_{i4}\,]$$

Utility gate loss (optional). Each cause gets an auxiliary classifier. The target q says which cause best explains the true emotion, so the gate is trained against the label.

$$q_{ik} = \frac{m_{ik}\,p_k(y_i \mid \tilde{f}_{ik})^{1/T}}{\sum_j m_{ij}\,p_j(y_i \mid \tilde{f}_{ij})^{1/T}} \qquad L_{total} = L_{task} + \lambda_{gate}\,\mathrm{KL}(q_i \,\|\, g_i) + \lambda_{aux}L_{aux}$$

Temperature scaling. One number T divides the logits [18]. It is fitted on validation data only, so it never changes which emotion is predicted.

$$\hat{p}_i = \mathrm{softmax}(l_i/T), \qquad T^* = \arg\min_T \sum_{i\in\mathrm{valid}} -\log \hat{p}_{i,y_i}$$

Seed ensembles and selective prediction. Averaging the calibrated outputs of several seeds reduces seed noise without retraining, and predictions with low calibrated confidence are flagged rather than trusted.

3.5 Merged systems

Approach A: merged per dataset

Each teammate merged CACG into their own pipeline and reported one dataset.

IEMOCAP: L1 + L3 (Divyansh Dubey). This system uses WavLM audio, VAD supervision, CACG with no gate loss, and temperature scaling, built on the authors' code. It was retrained from scratch on 20 seeds.

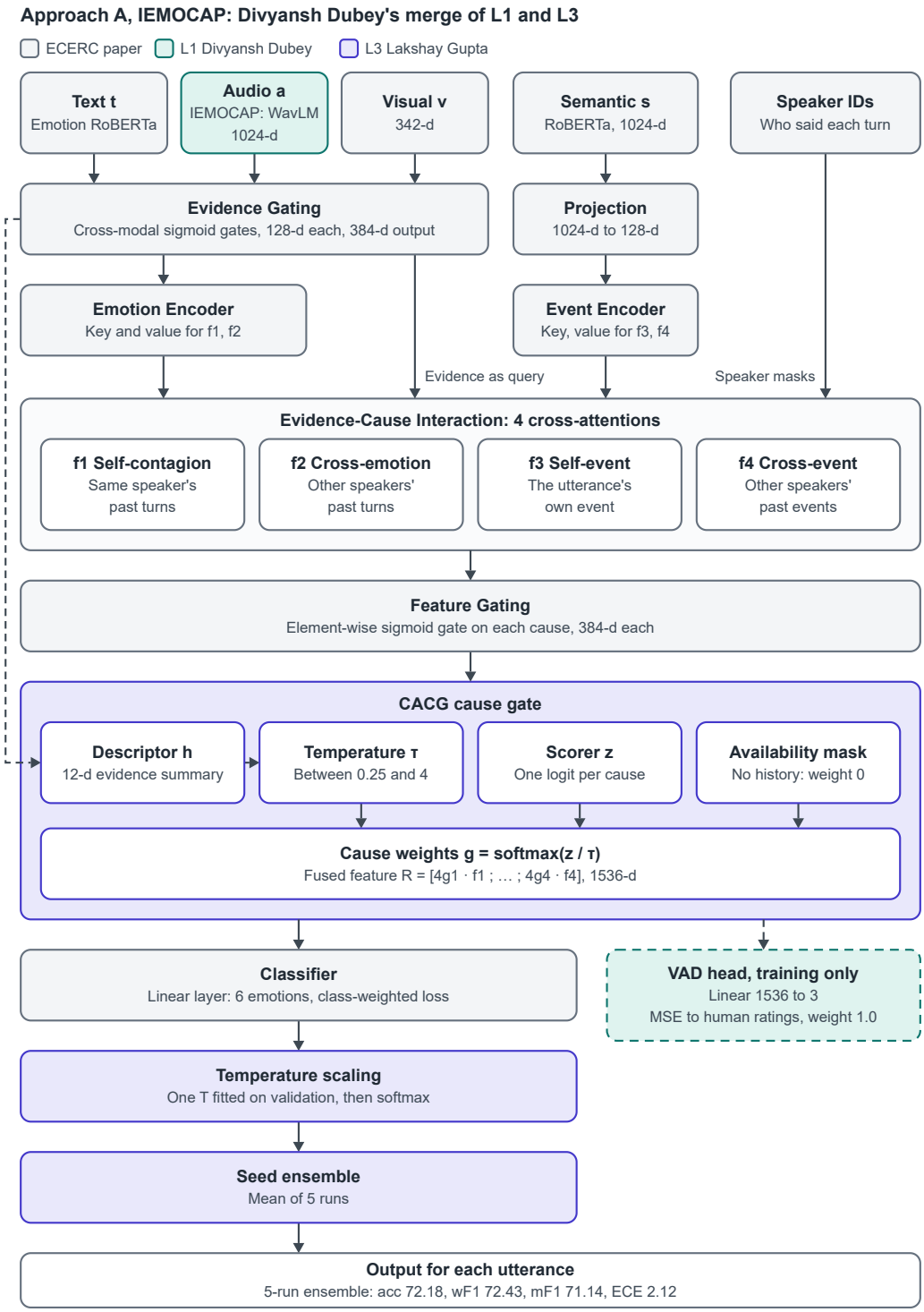


Figure 6: Approach A on IEMOCAP: WavLM, VAD head, CACG, temperature scaling and a 5-run ensemble

MELD: L2 + L3 (Keshav Mishra). This system uses CACG with no gate loss, the authors' focal loss, LLM fusion at $w = 0.2$, and temperature scaling. It was retrained on 3 seeds in `Merge-L2-L3/ext`, which is a copy of the rewritten code with L2 added, so unlike the IEMOCAP merge it does not run on the authors' code. Every w and T was fixed on dev before test was read.

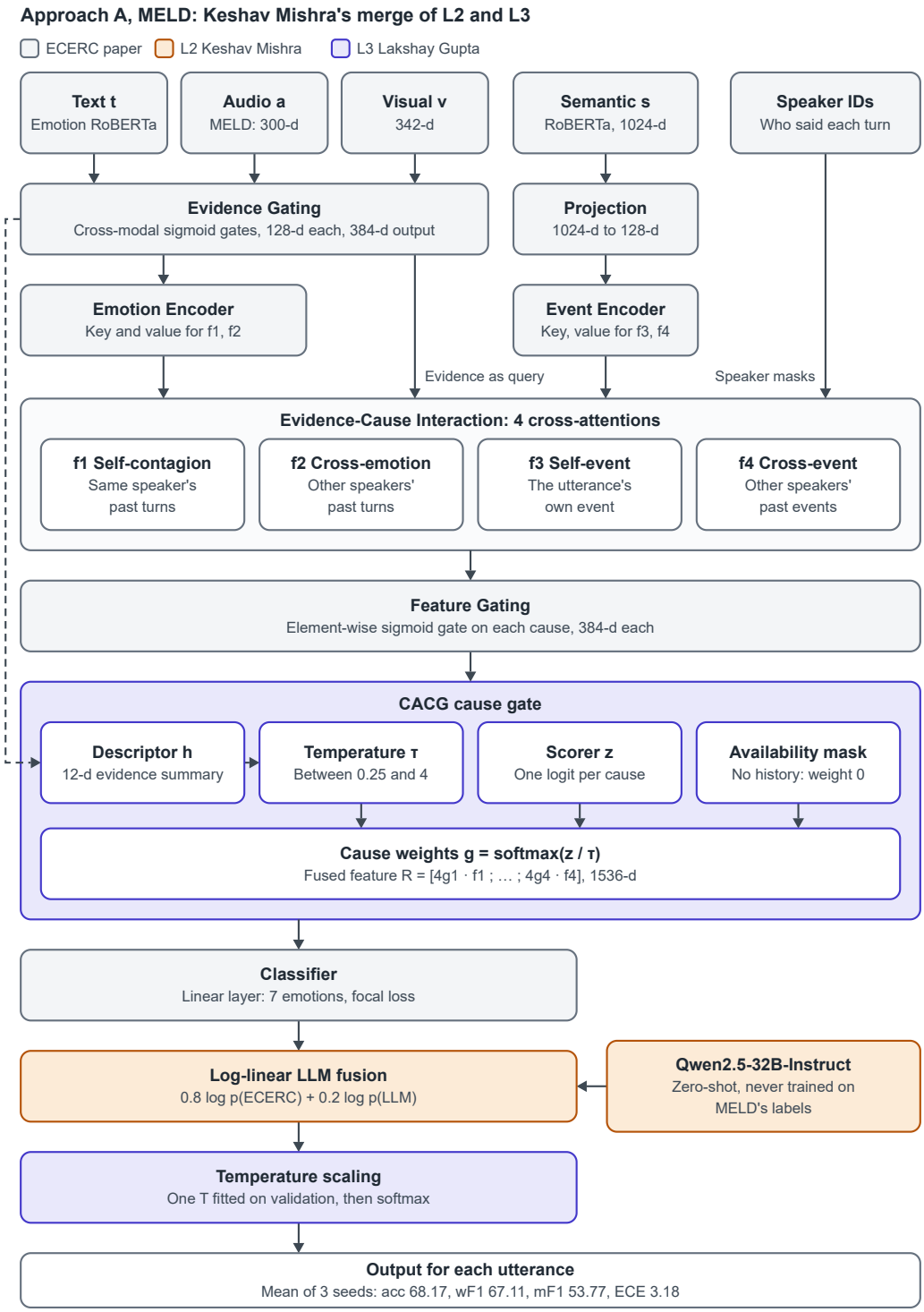


Figure 7: Approach A on MELD: CACG, LLM fusion and temperature scaling

Approach B: one rewritten codebase

All three limitations sit in one codebase, `Limitation-3/ECERC/ext/`, where every component is a command-line flag. One script trains any combination on either dataset, so each ablation row differs by flags alone. All runs were retrained on a laptop with a GeForce 940MX, and each dataset still has its own final configuration, as in the ECERC paper.

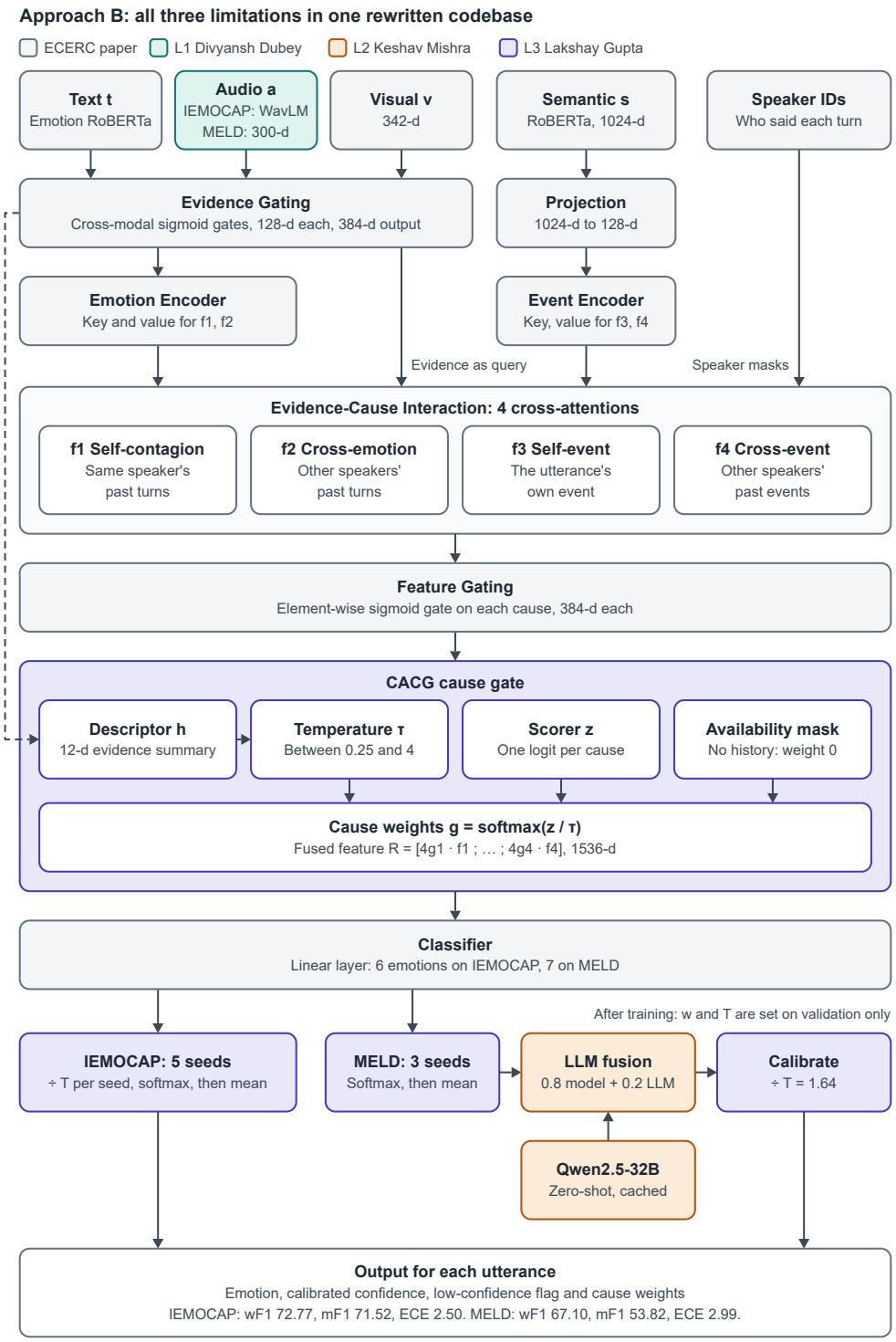


Figure 8: Approach B: one network, with a separate final route for each dataset

Component	From	IEMOCAP	MELD
Audio features	L1	WavLM	Release features; only IEMOCAP audio was extracted
VAD head + soft labels	L1	Tested, dropped	Not tested
CACG cause gate	L3	Dynamic τ, no gate loss	Dynamic τ, utility loss
Zero-shot LLM fusion	L2	No scores available	Yes, $w = 0.2$
Temperature scaling	L3	T per seed	$T = 1.64$ on the fused scores
Seed ensemble	L3	5 seeds	3 seeds

4 Experimental Setup

4.1 Datasets

IEMOCAP [2] uses Sessions 1 to 4 for training and Session 5 for testing, with 1,623 test utterances and 6 emotions. The first 10% of training conversations form the validation set. MELD [3] has 7 emotions:

MELD split	Dialogues	Utterances	Fear	Disgust
Train	1,038	9,989	268	271
Dev	114	1,109	40	22
Test	280	2,610	50	68

4.2 Metrics

Weighted F1 (wF1) is the paper's main metric, and macro-F1 (mF1) gives every emotion equal weight. Expected calibration error (ECE) is the gap between confidence and accuracy, in percent, and lower is better. TS means temperature scaling fitted on validation data. Pair errors count confusions between two similar emotions. Clear errors count utterances where every annotator voted for the gold label. On the rest, the annotators disagreed, so the correct label is debatable.

4.3 Evaluation protocol

- **Retrained baselines.** Retraining does not reach the authors' checkpoints (Section 5.2), so each limitation compares against a retrained baseline, and the checkpoints serve as a reference for overall accuracy.
- **L1.** All L1 runs used the authors' code and were compared with the same 20 seeds, paired seed by seed.
- **L2.** A "gate" is a preset minimum gain on dev data that a method needed before it could touch the test set. The gate was +0.5 mF1, and dev gains for the LLM were cross-fitted. Fusion was pre-registered, with a limit of −2.0 F1 on Joy set before the run.
- **L3.** The ablation grid used 3 seeds. CACG without a gate loss was rerun on 5 IEMOCAP seeds in the rewritten code, and on 20 seeds in the authors' code. The first MELD L3 runs were trained with focal $\gamma = 0$ before we found the release uses $\gamma = 1$.
- **Merged systems.** Approach A's IEMOCAP merge used 20 seeds and its MELD merge used 3. Approach B used 5 seeds on IEMOCAP and 3 on MELD. In every merged system, w and T were set on validation only.

4.4 Implementation of the merge

- **Code.** Approach A's IEMOCAP merge is in `Merge-L1-L3/`, its MELD merge is in `Merge-L2-L3/ext`, and Approach B is in `Limitation-3/ECERC/ext/`.
- **Hardware.** All Approach B runs were retrained on a laptop with a GeForce 940MX.
- **Data:** WavLM features, VAD values and annotator votes were aligned to all 7,433 IEMOCAP utterances, with none missing.
- **Fusion:** Keshav's cached LLM scores were fused with saved outputs, because the 32B LLM cannot run on the laptop. w and T were set on validation only.
- **Correction:** the first MELD grid used $\gamma = 0$, but the release uses $\gamma = 1$, as Keshav spotted. MELD was retrained at $\gamma = 1$, and all Approach B MELD figures use it.
- **Checks:** with new flags at their defaults, early epochs match earlier logs exactly, and the rewritten model still matches the official model with a maximum difference of 0.0.

5 Results and Analysis

5.1 Headline results

Dataset	Authors' checkpoints	Approach A	Approach B
IEMOCAP, ensemble wF1	71.78	72.43	72.77
IEMOCAP, ensemble mF1	70.82	71.14	71.52
MELD, wF1	66.46	67.11	67.10
MELD, mF1	50.87	53.77	53.82

5.2 Baseline reproduction

We reproduced the authors' results with their checkpoints, but no retraining reached them.

Source	Code	Runs	IEMOCAP acc	IEMOCAP wF1	MELD acc	MELD wF1
ECERC paper	Authors'	Mean of 3	71.60	71.78	67.32	66.46
Authors' checkpoints	Authors'	Mean of 3	71.60	71.78	67.32	66.46
One authors' checkpoint	Rewritten	1	71.60	71.77	67.55	66.58
Retrained, Divyansh Dubey	Authors'	20 seeds	69.50	69.70		
Retrained, Keshav Mishra	Authors'	3 seeds			66.87	66.00
Retrained, Keshav Mishra	Rewritten copy	3 seeds			66.83	65.72
Retrained, Lakshay Gupta	Rewritten	5 and 3 seeds	69.34	69.52	67.08	65.90

A blank cell means that person did not run that dataset.

Why retraining scores lower. IEMOCAP retraining lands about 2 wF1 below the checkpoints in both code bases, so the rewrite is not the cause. The authors say exact reproduction needs their environment. On IEMOCAP, seeds alone move wF1 by over 2 points (68.45 to 70.79 across 20 seeds). The released IEMOCAP checkpoints all score 71.77 to 71.78, so they are likely the best of several runs.

Because of this gap, each limitation compares against a retrained baseline, and the checkpoints serve as a reference for overall accuracy.

5.3 Limitation 1: category confusion (IEMOCAP)

Better audio worked: WavLM with VAD supervision raised wF1 by 1.43 and removed 20% of clear-cut Angry–Frustrated errors.

Metric (20 paired runs)	Baseline	WavLM + VAD	Change [95% CI]	Runs better
Accuracy	69.50	70.99	+1.48 [+1.06, +1.97]	19 of 20
wF1	69.70	71.13	+1.43 [+1.00, +1.87]	19 of 20
mF1	68.83	69.73	+0.90 [+0.47, +1.35]	15 of 20
ECE after TS	4.30	3.40	−0.90 [−1.33, −0.47]	16 of 20
Angry–Frustrated clear errors	16.9	13.5	−3.4 [−5.1, −1.7]	16 of 20
Happy–Excited clear errors	12.4	12.5	+0.1 [−1.2, +1.5]	7 of 20, 3 tied

WavLM alone gives +1.28 wF1, and VAD adds a further +0.15, which is within noise.

Happy–Excited stays unsolved. The two sound almost the same, and annotators drew the line between them inconsistently.

5.4 Limitation 2: class imbalance (MELD)

Zero-shot LLM fusion raised test macro-F1 by 3.38 on the authors' checkpoints (95% CI +0.64 to +5.73). The cross-fitted dev gain was +5.63 against a +0.5 gate, so fusion went to the test set once. It works because the LLM matches ECERC on macro-F1 with opposite errors: stronger on Disgust (37.58 against 31.43) and weaker on Joy (55.57 against 64.95).

System	Acc	wF1	mF1	Fear F1	Disgust F1	Joy F1
Authors' checkpoints	67.32	66.46	50.87	26.12	31.43	64.95
Authors' checkpoints + LLM	68.24	67.26	54.25	33.77	40.22	61.67
LLM alone	63.10	63.14	51.05	31.88	37.58	55.57
Retrained baseline	66.83	65.72	49.57	23.21	30.46	64.18
Retrained baseline + LLM	68.11	67.01	53.59	32.31	38.85	61.62

Fusion gained 4.02 mF1 on the retrained baseline (95% CI +1.55 to +6.45). The pre-registered verdict is a trade-off, not a clean success, because Joy F1 fell 3.28 points, past the −2.0 limit set before the run.

5.5 Limitation 3: cause gating and calibration (both datasets)

CACG did not change accuracy measurably, but it makes cause weighting readable and it slightly lowers calibration error. Temperature scaling gave the largest calibration gain, cutting IEMOCAP ECE from 14.36 to 3.72, and it works on any model.

Dataset and run	Model	Seeds	wF1	mF1	ECE raw	ECE + TS
IEMOCAP, authors' code	ECERC	20	69.70	68.83		4.30
IEMOCAP, authors' code	CACG	20	69.58			3.74
IEMOCAP, rewritten code	ECERC	5	69.52 ± 0.97	68.75	13.33	4.55
IEMOCAP, rewritten code	CACG, no gate loss	5	69.77 ± 1.28	69.06	14.36	3.72
MELD, rewritten code	ECERC	3	65.90 ± 0.28	49.78	3.62	2.92
MELD, rewritten code	CACG, utility loss	3	65.51 ± 0.08	49.25	3.97	3.20
MELD, rewritten copy (Keshav Mishra)	ECERC	3	65.72 ± 0.38	49.57	3.02	3.49
MELD, rewritten copy (Keshav Mishra)	CACG, no gate loss	3	65.96 ± 0.24	49.67	3.90	3.05

A blank cell means the source does not report that value. CACG changes wF1 by −0.39 to +0.25 across these runs, which is inside seed noise. The strongest calibration evidence is Divyansh's 20-seed rerun, where CACG lowered scaled ECE in 15 of 20 runs.

Interpretability of the cause weights

CACG shows which of the four causes the model relied on for each utterance, which ECERC cannot do directly. ECERC's gates act inside each cause vector, so they give no single score comparing one cause with another.

System	Self-contagion	Cross-emotion	Self-event	Cross-event
IEMOCAP, WavLM + CACG (Approach B)	0.50	0.38	0.08	0.05
IEMOCAP, WavLM + VAD + CACG (Approach A)	0.48	0.38	0.09	0.06
MELD, no gate loss, full history (Approach A)	0.34	0.42	0.14	0.10
MELD, utility loss, full history (Approach B)	0.28	0.27	0.23	0.23

- **Emotion causes dominate.** On IEMOCAP they carry 88% of the weight in the Approach B model.
- **Emotions differ.** Anger leans towards the speaker's own past (0.62 against 0.24), while Sad and Neutral draw about evenly on both speakers.
- **History matters.** Self-contagion grows from 0.39 with 1 to 2 prior utterances to 0.48 after more than 20. With no history, all weight goes to self-event.
- **The utility loss flattens the weights.** On MELD with full history, its gate entropy is 1.38 against a maximum of 1.39, so its weights say little. Without a gate loss, the weights stay readable.

These weights have clear limits. They show what the model relied on, not what truly caused an emotion, because we had no human cause labels to check them against. With full history, all seeds agree on the top cause for only 59.5% of IEMOCAP utterances. Gate entropy is almost the same on right and wrong predictions, so the weights do not signal errors. τ also stayed nearly constant in every variant without a gate loss.

5.6 Approach A: merged per dataset

Both Approach A systems beat the authors' checkpoints on overall score, and neither gain comes from CACG itself.

IEMOCAP: L1 + L3

Model	Acc	wF1	mF1	ECE raw	ECE + TS
Authors' checkpoint 1	71.60	71.77	70.73	11.12	2.52
Authors' checkpoint 2	71.53	71.78	70.70	11.07	2.95
Authors' checkpoint 3	71.66	71.78	71.02	11.30	2.97
Authors' checkpoints, average	71.60	71.78	70.82	11.16	2.81
Merged, single run (mean of 20)	71.36	71.59	70.19	13.64	3.41
Merged, 5-run ensemble	**72.18**	**72.43**	**71.14**	10.60	**2.12**

The ensemble row is the mean of 4 separate groups of 5 runs. It beat the authors' average in 3 of those 4 groups. Against the retrained baseline, the merged model gains 1.89 wF1 and is better in all 20 runs. A single run sits within 0.2 wF1 of the authors, 8 of the 20 runs beat them, and the best run reaches 72.95.

Clear-utterance errors	Baseline	WavLM + VAD	Merged (L1 + L3)
Angry–Frustrated	16.9	13.5 (better in 16 of 20)	15.4 (better in 12 of 20)
Happy–Excited	12.4	12.5	11.8 (better in 11 of 20)

On clear utterances, the merged model keeps part of the L1 gain on Angry–Frustrated, cutting errors by 1.5 against the baseline, but it gives back about half of what WavLM + VAD alone achieved. It is the only system that lowers Happy–Excited errors, though 11 of 20 runs is close to chance, so this is not a clear improvement.

MELD: L2 + L3

Row (3 seeds)	Acc	wF1	mF1	Fear F1	Disgust F1	Joy F1	ECE + TS	Dev mF1
Baseline	66.83	65.72	49.57	23.2	30.5	64.2	3.49	54.07
CB0.999	65.80	65.68	49.50	22.2	30.0	63.5	2.56	54.18
CACG	67.09	65.96	49.67	22.8	29.9	64.4	3.05	53.45
Baseline + LLM	68.11	67.01	53.59	32.3	38.9	61.6	3.38	59.96
CB0.999 + LLM	68.43	67.66	54.45	34.9	39.6	63.0	3.58	59.38
CACG + LLM (merged)	**68.17**	**67.11**	**53.77**	32.9	39.1	62.1	3.18	59.51

Merged system against the authors' checkpoints	Authors	Merged	Change	95% CI
Accuracy	67.32	68.17	+0.85	[−0.69, +2.38]
wF1	66.46	67.11	+0.65	[−0.94, +2.21]
mF1	**50.87**	**53.77**	**+2.90**	**[+0.42, +5.29]**

Only the macro-F1 interval excludes zero. Nearly all of the gain comes from the LLM: CACG given the LLM adds +0.18 mF1 [−0.24, +0.62], and the authors' checkpoints plus the LLM reach 54.25. CB0.999 + LLM has the best test score, but it ranked third on dev, so it was not selected.

Emotion F1	Neutral	Surprise	Fear	Sadness	Joy	Disgust	Anger
Authors' checkpoints	79.79	58.97	26.12	40.95	64.95	31.43	53.89
Merged system	79.62	61.22	32.88	46.14	62.05	39.06	55.42
Change	−0.17	+2.25	+6.76	+5.19	−2.90	+7.63	+1.53

5.7 Approach B and comparison of both approaches

Approach B's full results are its final rows in the ablations of Section 6.4. The two approaches reach almost the same results by the same route, so the conclusion does not depend on which code was used. On IEMOCAP, Approach B scores 0.34 wF1 higher, and on MELD the two are level, but neither gap is proven.

IEMOCAP	Runs	Acc	wF1	mF1	ECE + TS
Authors' checkpoints	3	71.60	71.78	70.82	2.81
Approach A, single run	20	71.36	71.59	70.19	3.41
Approach B, single run	5	71.77	71.92	70.65	2.89
Approach A, ensemble	5-run, 4 groups	72.18	72.43	71.14	2.12
Approach B, ensemble	5-run, 1 group	72.58	72.77	71.52	2.50

MELD	Runs	Acc	wF1	mF1	ECE + TS
Authors' checkpoints	3	67.32	66.46	50.87	3.02
Authors' checkpoints + LLM	3	68.24	67.26	54.25	
Approach A, mean over seeds	3	68.17	67.11	53.77	3.18
Approach B, mean over seeds	3	68.28	67.23	54.10	2.99
Approach B, ensemble	3	68.16	67.10	53.82	2.99

6 Ablation Study

This section records every method tried for each limitation, then the component ablations of the unified codebase.

6.1 Limitation 1: every method tried

Five training changes on the original features failed to fix the confused pairs, which showed the real problem was missing information. All runs used the authors' code.

#	Method	Why it was tried	Result	Verdict	Why
1	Confusion-aware contrastive loss	Separate the internal vectors of confused emotions	Looked like about −15 pair errors on 5 seeds; 174.3 against 173.9 on 10 seeds, wF1 69.63 against 69.79	Failed	The classes are not separated even on training data, and the 5-seed gain was noise
2	Pair penalty	Punish probability on the partner class	Same 10-seed result, alone or combined	Failed	It can only move the boundary. Even a shift tuned on test added just 1.2 wF1, and tuned on validation it lost 1.5

#	Method	Why it was tried	Result	Verdict	Why
3	Ensembling	Cancel random errors	wF1 69.70 to 70.26 and ECE 12.0 to 10.0 with 5 models, but pair errors stayed	Helped overall, not the pairs	Different runs get the same utterances wrong
4	VAD supervision, original audio	The pairs differ in energy and dominance	Angry–Frustrated errors 97.9 to 95.9 (15 of 20 runs); wF1 unchanged at 69.78	Small real gain	It learned arousal (correlation 0.73), but the old audio holds little about how something is said
5	Soft labels from annotator votes	Stop punishing agreement with minority annotators	ECE 12.1 to 6.1, NLL 0.84 to 0.74, wF1 69.82, Happy–Excited errors up about 5	Calibration only; dropped	The model learned to hedge, and hedges tipped towards the larger class. Temperature scaling calibrates without this cost
6	WavLM audio + VAD supervision	Add new information to the model	wF1 +1.43, Angry–Frustrated clear errors down 20%	Final method	Modern speech features carry what the old ones lacked

6.2 Limitation 2: every method tried

Each reason below is an interpretation of the results, not a proven cause.

Group	Method	Result	Verdict	Likely reason
Loss and sampling	Class-balanced loss (CB0.999)	Test mF1 49.52 against 50.40	Failed	Fear false positives rose from 25 to 95 per run; the extra Fear predictions were only 9% correct
Loss and sampling	Stronger focal loss ($\gamma = 2$)	50.06	Failed	The authors' loss is already focal, so this is a small change
Loss and sampling	Oversampling Fear and Disgust dialogues	50.02	Failed	It repeats the same few examples and adds no new evidence
Loss and sampling	Focal $\gamma = 2$ + oversampling	49.13, the lowest	Failed	It combines both weaknesses
Decision rule	Post-hoc logit adjustment	−0.24 [−1.06, +0.68]	Failed	It moved mostly correct predictions: 19 extra Fear predictions gave 1 correct one
Representation	Mean of 4 RoBERTa layers	Dev −0.07	Failed the gate	Fear and Disgust stayed at 30 to 36 dev F1 under every candidate
Representation	Confusion-aware contrastive loss	Dev −0.22	Failed the gate	Overlapping emotions stay hard to separate, whatever the loss
Representation	Auxiliary sentiment head	Dev +0.19	Failed the gate	Below the +0.5 needed to reach test
Representation	Learning rate 3e-5	Dev −0.06	Failed the gate	It peaked 20 epochs earlier then overfit, so under-training is not the cause
Trained expert	RoBERTa-large text expert	+0.27 [−1.92, +2.29]	Inconclusive	Fear rose 3.40, but Disgust and Joy fell
Trained expert	Qwen2.5-7B LLM, fine-tuned	Cross-fitted dev −0.98	Failed the gate	It learned MELD's skew, and reached only 22 to 23 dev Disgust F1
Better evidence	Whisper-large-v3 audio	Dev +0.08	Failed the gate	Text dominates on MELD; better audio moved Sadness up and Joy down
Final	**Zero-shot LLM fusion**	**+3.38 [+0.64, +5.73]**	**Worked, with a trade-off**	It was never trained on MELD's skewed labels, so its errors differ from ECERC's

Group	Method	Result	Verdict	Likely reason
After final	Distilling the LLM into ECERC	Cross-fitted dev −0.42	Failed the gate	The LLM is still needed at inference

The first group's test baseline is 50.40 mF1. Every variant there was below it on all three seeds. A "gate" is a preset minimum gain on dev data that a method needed before it could touch the test set.

6.3 Limitation 3: every method tried

#	Method	Why it was tried	Result	Verdict	Why, and what came next
1	Empty-history attention fix	Stop attention spreading over future turns	IEMOCAP wF1 70.11 to 68.99; MELD 65.39 to 65.36	Failed	It adds a fake history signal; replaced by the mask
2	Masked cause softmax, fixed τ	Create the missing cause distribution	IEMOCAP 69.78, MELD 65.32	Neutral	It gives cause weights but no accuracy; moved to per-utterance τ
3	Dynamic τ, no gate loss	Let weak evidence soften the weights	IEMOCAP 69.77 on 5 seeds; τ spread only 0.008	Kept	Nothing in training rewards τ for varying, so it stays nearly fixed
4	Proposal's entropy loss	Train the gate as the proposal planned	IEMOCAP 69.36; MELD 65.20; Disgust F1 30.2 to 21.0	Failed	Its target comes from the same inputs as τ and never checks the label; replaced by the utility loss
5	Utility loss, $T = 1$	Ground the gate target in the label	MELD 65.51 against 65.90 at $\gamma = 1$; flattens the weights	No gain	The small gain seen at $\gamma = 0$ did not hold
6	Utility loss, $T = 0.25$	Test a sharper target	IEMOCAP 69.52, MELD 65.50	No gain	Dropped
7	Global temperature scaling	Fix over-confidence directly	IEMOCAP ECE 14.36 to 3.72, predictions unchanged	Worked	Kept in every final system
8	Cause-aware temperature scaling	Set T per utterance from gate signals	4.11 against 2.95 on IEMOCAP	Failed	Over-confidence is about the same at every context length, so one T already fits
9	Availability mask	Remove causes with no history	Applies to 5.2% of IEMOCAP and 33.7% of MELD test utterances	Kept	Removes noisy vectors; effect not measured alone
10	Seed ensemble	Reduce seed noise without retraining	IEMOCAP CACG 69.77 to 70.68 wF1	Worked	Kept
11	Selective prediction	Hold back risky answers	Accuracy about 70% to 84.5% on the most confident half	Worked	Became the demo's low-confidence flag

Rows 1, 2, 4, 6 and 8 used 3 seeds, and their MELD figures were trained at $\gamma = 0$ before we found the release uses $\gamma = 1$. These explanations are our interpretation, and we did not test them separately.

Ablation grid, 3 seeds	IEMOCAP wF1	IEMOCAP ECE + TS	MELD wF1 ($\gamma = 0$)	MELD ECE + TS ($\gamma = 0$)
ECERC	70.11 ± 0.46	4.21	65.39 ± 0.34	3.01
ECERC + attention fix	68.99 ± 1.36	5.66	65.36 ± 0.22	3.18
Masked softmax, fixed τ	69.78 ± 1.11	3.87	65.32 ± 0.15	2.41
Dynamic τ, no gate loss	70.65 ± 0.86	2.95	65.34 ± 0.16	2.42
Dynamic τ + entropy loss	69.36 ± 1.29	4.26	65.20 ± 0.18	2.54
Dynamic τ + utility loss, $T = 1$	69.44 ± 1.03	4.22	65.59 ± 0.04	3.18
Dynamic τ + utility loss, $T = 0.25$	69.52 ± 0.32	3.55	65.50 ± 0.13	2.91

The +0.54 wF1 for dynamic τ on 3 IEMOCAP seeds shrank to +0.25 on 5 seeds and −0.12 on 20, so

these 3-seed differences should not be read as gains.

6.4 Approach B: component ablations

Each row below differs from the others by command-line flags alone.

IEMOCAP

Variant	Seeds	wF1, single	mF1, single	ECE + TS	Ensemble wF1	Ensemble mF1	Ensemble ECE
ECERC retrained	5	69.52 ± 0.97	68.75	4.55	70.29	69.45	2.40
+ L3 CACG	5	69.77 ± 1.28	69.06	3.72	70.68	70.14	2.79
+ L1 VAD + soft labels, IS10 audio	3	70.07 ± 0.24	69.08	3.76	70.07	69.09	3.12
+ L1 WavLM	5	71.60 ± 0.32	70.10	2.94	71.92	70.30	2.28
+ L1 WavLM + VAD + soft labels	5	70.93 ± 0.87	69.44	3.05	71.78	70.28	2.50
+ L1 WavLM + L3 CACG (final)	5	**71.92 ± 0.25**	**70.65**	2.89	**72.77**	**71.52**	2.50
+ L1 WavLM + VAD + soft labels + L3	5	71.46 ± 0.44	69.96	2.52	72.66	71.03	2.81

WavLM gave the largest gain and cut the seed spread from 0.97 to 0.32. VAD and soft labels together cost about 0.5 to 0.7 wF1 on top of WavLM, so they were dropped. This differs from Approach A, where soft labels were dropped first and VAD was kept. Approach B never tested VAD on its own with WavLM.

MELD

Variant (3 seeds, $\gamma = 1$)	wF1, single	mF1, single	ECE raw	ECE + TS	Ensemble wF1	Ensemble mF1	Ensemble ECE
ECERC retrained	65.90 ± 0.28	49.78	3.62	2.92	65.99	50.05	2.73
+ L3 CACG, utility loss	65.51 ± 0.08	49.25	3.97	3.20	65.83	49.86	3.90
+ L2 LLM fusion	67.01 ± 0.22	53.64	16.81	3.05	67.31	54.29	3.22
+ L2 LLM fusion + L3 CACG (final)	**67.23 ± 0.21**	**54.10**	14.98	2.99	**67.10**	**53.82**	2.99

Fusion raised raw ECE from 3.6 to 16.8, and temperature scaling brought it back to 3.1, so the fused confidence cannot be trusted without scaling. The plain ECERC + LLM ensemble is slightly above the final one, within noise.

7 Discussion

7.1 What both approaches agree on

- **New evidence beat new losses.** WavLM on IEMOCAP and the zero-shot LLM on MELD gave the gains, after loss and sampling changes had failed in both limitations.
- **CACG adds no proven accuracy.** Its value is readable cause weights and a small calibration gain.
- **Temperature scaling is needed.** It keeps confidence honest, above all after LLM fusion.
- **Calibration ties with the authors once scaled.** After the same scaling, the authors' checkpoints reach 2.81 and 3.02, about level with our final systems. Most of the drop from raw ECE comes from scaling, not from our models.

7.2 Where the approaches differ

Choice	Approach A	Approach B	What it means
IEMOCAP code	Authors' code	Rewritten code	Same model, so the code is not the main difference
MELD code	Copy of the rewritten code	Rewritten code	Both MELD systems use the same core model code
IEMOCAP loss	WavLM + VAD	WavLM only	VAD adds +0.15 wF1 in Approach A, within noise
MELD gate loss	None	Utility loss	Scores are level, but weights are only readable without the loss
IEMOCAP seeds	20	5	Approach A's evidence is much stronger
Model selection	Validation, with VAD kept after comparing scores	Test ablation for the IEMOCAP choice	Approach B's IEMOCAP result is slightly optimistic
Coverage	One dataset per system	Both datasets, with a demo	Approach B is simpler to run and show

7.3 Final approach

The final method is one recipe with a different evidence source per dataset. On IEMOCAP, it is ECERC with WavLM audio, CACG, temperature scaling and a 5-seed ensemble. On MELD, it is ECERC with CACG, zero-shot LLM fusion and temperature scaling, and CACG should run without the utility loss, so its weights stay readable.

For headline numbers, we recommend quoting Approach A's 72.43 wF1 on IEMOCAP as the best-evidenced result, and Approach B's 72.77 as the matching result in the unified code.

7.4 Demo interface

A local page replays the test conversations of both final models. Each utterance shows the predicted emotion, calibrated confidence, a low-confidence flag and the cause weights. Flagged predictions are right 47.0% of the time on IEMOCAP and 37.7% on MELD, against 78.7% and 76.3% for the rest. It cannot take typed sentences, because the final models need raw audio features, emotion-tuned RoBERTa weights we do not have, and the 32B LLM.

8 Limitations

The gains are real but modest in certainty, and several caveats apply to how they should be read.

- **Repeated test use.** The MELD test set has been used four or more times. After correcting for four looks, the interval on the +2.90 mF1 gain is [−0.30, +5.96], which includes zero.
- **Selection.** Approach B chose its IEMOCAP model from test scores, because validation preferred the original IS10 audio. On MELD, dev ranks ECERC + LLM above the merged system, which was named because it combines both limitations.
- **Seeds.** Approach B used 5 and 3 seeds, so small differences cannot be confirmed. Only Approach A's IEMOCAP study has 20 paired seeds.
- **Small classes.** Test has only 50 Fear and 68 Disgust utterances, so per-emotion changes are noisy.

- **Baseline variation.** MELD retraining gave 65.72 to 66.00 wF1 across three runs of the same recipe, and the cause is not established.
- **Inference cost.** The MELD system needs a 32B LLM, used here through cached scores. WavLM needs raw audio.
- **LLM exposure.** The LLM's pretraining data may include Friends transcripts or MELD's public files. Alone it is weaker (63.14 wF1), which argues against memorised labels.
- **No cross-dataset tests.** WavLM was not tested on MELD, and LLM fusion was not tested on IEMOCAP.
- **Code availability.** Approach A's IEMOCAP merge code is in `Merge-L1-L3/`, but its run outputs are not in the repository, so its figures come from the written report.

9 Conclusion and Future Work

9.1 Conclusion

We extended ECERC on three limitations and beat the authors' checkpoints on both datasets, in two separate ways. On IEMOCAP, the final systems reach 72.43 (Approach A) and 72.77 (Approach B) ensemble wF1 against 71.78. On MELD, they reach 53.77 and 53.82 mF1 against 50.87. The clearest lesson is that new evidence beat new losses: 17 methods that changed the loss, sampling, decision rule or training budget fell short across Limitations 1 and 2, while WavLM audio and zero-shot LLM fusion succeeded. CACG adds no proven accuracy, but it makes cause weighting readable, and temperature scaling keeps the final systems' confidence honest. The two approaches agree on all of this, and the gaps between them are too small to rank them.

9.2 Future work

Each item follows from a limit or an open question above.

- **Cross-dataset tests.** Extract WavLM features for MELD, and LLM scores for IEMOCAP, to test whether each evidence source helps on the other dataset.
- **Happy–Excited confusion.** This pair stays unsolved by every method. Because annotators drew the line between them inconsistently, label schemes that model annotator disagreement are a natural next step.
- **Cheaper inference.** Distilling the LLM into ECERC failed the dev gate (-0.42), so the MELD system still needs a 32B LLM at inference. Smaller LLMs or other distillation targets could remove that cost.
- **Validated cause weights.** Collect human cause labels to check whether CACG's weights match what truly caused an emotion, and add a training signal that rewards τ for varying, since it stayed nearly constant without a gate loss.
- **Stronger evidence for Approach B.** Rerun it on 20 paired seeds, select the IEMOCAP model on validation, test VAD on its own with WavLM, and evaluate on data not used for earlier comparisons, given the repeated use of the MELD test set.
- **Typed input in the demo.** Supporting typed sentences needs a text-only path that does not depend on raw audio features, emotion-tuned RoBERTa weights or the 32B LLM.

References

[1] T. Zhang and Z. Tan. ECERC: Evidence-Cause Attention Network for Multi-Modal Emotion Recognition in Conversation. In *Proceedings of ACL*, 2025.

[2] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee and S. S. Narayanan. IEMOCAP: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4):335–359, 2008.

[3] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria and R. Mihalcea. MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations. In *Proceedings of ACL*, 2019.

[4] N. Majumder, S. Poria, D. Hazarika, R. Mihalcea, A. Gelbukh and E. Cambria. DialogueRNN: An Attentive RNN for Emotion Detection in Conversations. In *Proceedings of AAAI*, 2019.

[5] D. Ghosal, N. Majumder, S. Poria, N. Chhaya and A. Gelbukh. DialogueGCN: A Graph Convolutional Neural Network for Emotion Recognition in Conversation. In *Proceedings of EMNLP-IJCNLP*, 2019.

[6] J. Hu, Y. Liu, J. Zhao and Q. Jin. MMGCN: Multimodal Fusion via Deep Graph Convolution Network for Emotion Recognition in Conversation. In *Proceedings of ACL-IJCNLP*, 2021.

[7] B. Schuller, S. Steidl, A. Batliner, F. Burkhardt, L. Devillers, C. Müller and S. Narayanan. The INTERSPEECH 2010 Paralinguistic Challenge. In *Proceedings of Interspeech*, 2010.

[8] S. Chen et al. WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518, 2022.

[9] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey and I. Sutskever. Robust Speech Recognition via Large-Scale Weak Supervision. In *Proceedings of ICML*, 2023.

[10] Y. Liu et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692, 2019.

[11] J. C. Peterson, R. M. Battleday, T. L. Griffiths and O. Russakovsky. Human Uncertainty Makes Classification More Robust. In *Proceedings of ICCV*, 2019.

[12] T.-Y. Lin, P. Goyal, R. Girshick, K. He and P. Dollár. Focal Loss for Dense Object Detection. In *Proceedings of ICCV*, 2017.

[13] Y. Cui, M. Jia, T.-Y. Lin, Y. Song and S. Belongie. Class-Balanced Loss Based on Effective Number of Samples. In *Proceedings of CVPR*, 2019.

[14] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit and S. Kumar. Long-tail learning via logit adjustment. In *Proceedings of ICLR*, 2021.

[15] P. Khosla et al. Supervised Contrastive Learning. In *Advances in NeurIPS*, 2020.

[16] Qwen Team. Qwen2.5 Technical Report. arXiv:2412.15115, 2024.

[17] G. Hinton, O. Vinyals and J. Dean. Distilling the Knowledge in a Neural Network. arXiv:1503.02531, 2015.

[18] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger. On Calibration of Modern Neural Networks. In *Proceedings of ICML*, 2017.

[19] M. P. Naeini, G. F. Cooper and M. Hauskrecht. Obtaining Well Calibrated Probabilities Using Bayesian Binning. In *Proceedings of AAAI*, 2015.

[20] B. Lakshminarayanan, A. Pritzel and C. Blundell. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in NeurIPS*, 2017.

[21] Y. Geifman and R. El-Yaniv. Selective Classification for Deep Neural Networks. In *Advances in NeurIPS*, 2017.

Project sources. The figures in this report come from the following team documents and files.

- Limitation 1 and IEMOCAP merge: `ECERC Revisited on IEMOCAP Final Report (Limitations 1 and 3).pdf`, `Merge-L1-L3/README.md`.
- Limitation 2 and MELD merge: `Merge-L2-L3/REPORT.md`, `Merge-L2-L3/README.md`, `Limitation-2/README.md`.
- Limitation 3 and Approach B: `FINAL_MODEL.md`, `Limitation-3/ECERC/runs/RESULTS.md`, `Limitation-3/ECERC/runs/RESULTS_merged.md`.
- Architecture diagrams: the SVG files in `architecture/`, generated by `architecture/make_svgs.py`.

A Individual contributions

Each member owned one limitation, and all three reproduced ECERC on their datasets.

Member	Main work	Key result
Divyansh Dubey (12340690)	L1 on IEMOCAP: six methods, WavLM extraction for 7,433 utterances, VAD and vote labels, 20-seed paired study, L1 + L3 merge on the authors' code, IEMOCAP final report	wF1 +1.43; 20% fewer clear-cut Angry–Frustrated errors; ensemble 72.43 wF1
Keshav Mishra (12341140)	L2 on MELD: 12 methods, pre-registered zero-shot LLM fusion, L2 + L3 merge, MELD final report and architecture figures. Spotted the $\gamma = 1$ loss	mF1 +3.38 from fusion; merged 53.77 mF1, +2.90 over the authors
Lakshay Gupta (12341300)	L3 on both datasets, plus the rewritten code and the all-in-one merge (details below)	CACG, ECE 14.36 to 3.72 with scaling, ensemble 72.77 wF1 and 53.82 mF1

Lakshay Gupta's work in detail:

- **Unified code:** rewrote ECERC as one package for both datasets, with a maximum output difference of 0.0 from the official model. Teammates' MELD merge was built on it.
- **Code audit:** found the missing cause softmax, the empty-history attention problem and the MELD modality routing.
- **CACG:** designed the masked cause distribution, 12-d descriptor, per-utterance temperature and availability mask.
- **Utility gate loss:** designed a label-grounded replacement for the proposal's entropy loss.
- **Interpretability:** exposed per-utterance cause weights, and showed which causes the model relies on for each emotion.
- **Calibration:** added ECE, NLL and reliability analysis, temperature scaling and cause-aware scaling.
- **Seed ensembles and selective prediction:** turned confidence into a low-confidence flag.
- **Ablation:** ran the 42-run L3 grid and the merged ablation on both datasets.
- **Approach B:** merged all three limitations behind flags, retrained on one laptop, and built the demo interface.

- **Documentation:** wrote `FINAL_MODEL.md`, a merging guide for teammates, the SVG architecture diagrams and this combined report.